

ARCHITECTURE & ENGINEERING TRACK

The Agentic Banking Experience

Reimagining consumer banking around an AI agent that reasons and acts on a customer's money.

TRACK	TIMELINE	AUDIENCE	DELIVERABLES
Architecture & Engineering	5 working days	Business & technical	3 parts

01 Context

A large Nordic bank wants to reimagine its consumer banking app. Today, customers navigate menus and trigger every action themselves. We want to explore what happens when an AI agent becomes the primary interface — one that knows the customer's financial history and goals, acts proactively, and shifts the customer from operator to overseer.

A prototype is easy to imagine. Making it *real* — accurate, safe, and operating on actual money inside a regulated bank — is the interesting part, and that's what this track is about.

Your task: envision what this experience looks like, **and outline the system that would make it work in production.**

02 The core questions

- Q1 *When an AI agent handles most of the work, what does the human actually need to see, do, and decide?*
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- Q2 *What does it take to build an agent that can be trusted to reason and act on a customer's money — and how would you architect it?*
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The second question is the heart of this track. We're most interested in how you think about building GenAI systems — not just how they feel to use. We care less about pixel-perfect UI than about clear, honest thinking: a modest prototype with a sharp design behind it beats a polished demo that hand-waves the hard parts.

03 What to build

Three parts. Don't over-scope any of them — we're looking for clear thinking, not volume.

Part A A working prototype

Build something we can open and play with. At least one agent flow should feel real. Wire in a real model if you like, or simulate it with mock data — either is fine, as long as your writeup explains how the real thing would work. Keep it easy to run: open in a browser, or a short README, minimal setup.

Part B Architecture & system design

A short written design, with diagrams, for the production system behind your prototype. This is the centre of the track — the document you'd show a bank's engineering leads to say "here's how we'd actually build it." Show us the system; don't just describe it.

Part C Process & decisions

A short note on how you worked:

- The tools, frameworks, and models you used, and why
- Where AI assistance helped — and where it didn't
- Your main design decisions and the thinking behind them
- What you'd build first for production, and what you left out for a 5-day exercise

Deliver Parts B and C as **both** a Markdown source and a clean, reader-friendly HTML version — something pleasant to open in a browser, no build step. Both should contain the same content.

05 What we'd love you to cover

Cover the themes below in whatever depth and order makes sense to you. There's no checklist to tick and no single right answer — these are simply the things we'll be curious about as we read. Pick the ones you find most interesting and go deep there.

Interaction & autonomy

How does the customer work with the agent? When does it act on its own, when does it ask first, and how does that change with risk — a small transfer isn't a mortgage.

Reference architecture — the core

The end-to-end system: the main components, how data and control flow between them, where the models sit, and the line between your AI system and the bank's existing systems of record. A diagram or two goes a long way here.

Grounding in the customer's data

How the agent gets accurate, current access to the customer's history, products, and the bank's policies — without inventing facts.

Taking action safely

How the agent discovers what it's allowed to do and carries it out in a safe, auditable way — and how you'd add a new capability later without rewriting everything.

Breaking the problem down

How you'd split the system into specialised parts that coordinate — and what owns planning, execution, and oversight.

Reliability & safety

How you keep it from doing the wrong thing, how you'd know it's working, and what happens when something fails — latency, cost, outages, confident-but-wrong answers.

Enterprise & regulatory reality

How the design holds up in a real bank: auditability, privacy, authentication for agent-initiated actions, and shared or joint accounts.

There's no required tech list. When you choose a pattern or tool, a sentence on *why* tells us more than the name itself.

06 Practical details

Use whatever stack and tools you're comfortable with — no restrictions on languages, frameworks, models, or AI assistance. Assume a mixed audience of business stakeholders and our technical team, so your work should make sense to both.

Send all three parts together — a folder, repo, or deployed URL we can open locally. The main thing: we can run the prototype and open the written parts without a build step or a scavenger hunt.

IN SHORT

Timeline: 5 working days. There's no single right answer — we're interested in how you think about the problem and the system behind it. If something is unclear or you'd scope it differently, tell us; that's part of the signal.